MILITARY SPECIFICATION FOR DIAPHRAGM ASSEMBLY

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification states the requirements for a diaphragm assembly used to seal the top of a missile launcher tube,

2. APPLICABLE DOCUMENTS

2.1 Documents - The following documents, to the extent specified herein, of the issue in effect on the date of Invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-MD-21524	Detonator Assembly
MIL-E-5272	Environment Testing, Aeronautical' and Associated Equipment, General Specification for
Code Ident 10001 OS 10089A	Metal-Clad 'Igniter Fuse

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

COMMERCIAL

American Table of Distances

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DRAWINGS

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS, Code Ident 10001

BuWeps LD 419636	Diaphragm Assembly
SA-2440495	Shock Test Fixture
SA-2586826	Bilateral Test Fixture
EM-2586865	Bilateral Test Control Panel
SA-2586881	Submersion Fixture

PUBLICATIONS

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS, Code Ident 10001

NAVWEPS OP 400	General Instructions for Design, Manufacture, and Inspection of Naval Ordnance Equipment
NAVORD INST 4355.1	Equipment for Use by Government Inspectors in Final Inspection of Naval Ordnance Items
BUWEPS INST 4355.5	Bureau of Naval Weapons Calibration Program
Standards Laboratory Information Manual	

ARMED SERVICES

Armed Services Explosives Safety Board

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other Publications - The following documents form a part of this specification, Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standards for Federal Services

CODE OF FEDERAL REGULATIONS

49 CFR 71-78	Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles

(Applications for copies should be addressed to the Superintendent of Documents. Government Printing Office, Washington 25, D.C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois)

3. REQUIREMENTS

3.1 Description - The diaphragm assembly consists of a laminated Mylar membrane, an upper and lower clamping ring, the igniter fuse’(OS 10089 a Protective tunnel, Mylar tape, four detonator bases$ and the necessary Bolts. The diaphragm assembly seals the upper end of the launcher tube from sea water entry and permits a controlled atmosphere to be maintained within the tube. At the instant of missile launch, the diaphragm is cut by the igniter fuse which is actuated by four detonators (MIL-D-21524).

3.2 General Requirements

3.2.2 Preproduction Samples - Unless otherwise specified in the contract or order, preproduction samples of the diaphragm assembly shall be manufactured using the methods and” procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification (see paragraph 4.2.1). At the discretion of the procuring agency. the requirement for a preproduction sample

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may be waived and shall then be satisfied by substitution of a production unit chosen at random by the Government Inspector.

3.2.2 Materials - Unless otherwise specified in the contract, requisition, or order, all materials and pup. chased parts entering into the final assembly of the diaphragm assembly shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BuWeps LD 419636,

3.2.3 Finishes, Coatings, Construction, and Dimensions Finishes, coatings, construction and dimensions for the diaphragm assembly shall be in accordance with the applicable drawings, specifications, and publications listed in BuWeps LD 419636.

3.2.4 Marking - Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in BuWeps LD 419636.

3.2.5 Interchangeability - Individual assemblies" and piece parts of the assemblies fabricated in agreement with their applicable drawings and specifications shall be interchangeable with like numbered parts.

3.2.6 Gages - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, requisition, or order, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractors inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.7 Threads - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

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3.2.8 Requests for Deviation - Requests for deviation from this specification or any specification, publication, material, or process specified herein, shall be as set forth in NAVWEPS OP 400.

3.2.9 Conflicting Requirements - In event of conflict between this specification and any specification, publication or drawing referenced herein, this specification shall govern.

3.2.10 Tolerances - Unless otherwise specifically noted, tolerances and accuracies stated herein are absolute.

3.3 Performance Requirements

3.3.1 Unilateral Flexure-; With 6 inches of water above the membrane, the diaphragm assembly shall withstand 6 psig air pressure applied below the membrane for 15 minutes. The membrane shall deflect at least 1 inch, but not more than 4 inches, shall show no leakage in excess of two 1/8-inch diameter air bubbles per minute, shall show no permanent deformation in excess of 2-1/2 inches from the nominally flat surface, and thereafter shall be capable of meeting all performance requirements of this specification.

3.3.2 Leakage, Fuse Assembly - The tunnel and tapes under 6 inches of water, shall withstand an internal air pressure equivalent to 20-25 inches of water for 10 minutes without leakage or damage to the tunnel or tape, and thereafter shall be capable of meeting all performance requirements of this specification.

3.3.3 Static Pressure - With 6 inches of water above the membranes the diaphragm assembly shall withtand an air pressure of 30 psig, applied downward for 15 minutes. During such pressure, the membrane may deflect to any extent but shall show no water leakage, and thereafter shall show no tear at any clamping hole nor any sign of rupture.

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3.3.4 Bilateral Flexure - with 6 inches of water above the membrane and with an air pressure of 30 psig maintained below the membrane, the diaphragm assembly shall withstand air-pressure cycles of 25 to 40 psig applied above the diaphragm at the rate of 1 cycle-per-minute for 50 cycles. The diaphragm assembly shall show no water leakage and no permanent deformation of the membrane in excess of 3 inches from the, nominally flat surface, and thereafter shall be capable of, meeting all performance requirements of this specification.

3.3.5 Rupture - The membrance, with 6 inches of water above and 5 psig air pressure below, shall be cut to the igniter fuse pattern when the igniter fuse is detonated by any one detonator.

Environmental Conditions

3.4.1 Vibration, Shipping Condition - The diaphragm, packaged and packed in accordance with section 5 herein, shall withstand the following vibration conditions, shall show no damage, and thereafter shall be capable of meeting all-performance requirements of this specification.

VIBRATION CRITERIA (MAX) FOR PACKAGING

Type of TRANSPORTATION	vertical cps	vertical g	Longitudineinal cps	Longitudineinal g	Lateral cps	Lateral g
Truck	200	2.5	150	1.5	200	1.0
	10	1.0	10	1.0	10	1.0
Rail	60	2.0	60	1.0	30	1.0
	10	1.0	10	0.5	10	0.5
Aircraft	60-200	1.0	60-100	0.5	60	1.0
	100-200	0.5			100	0.5
Ship (surface)	10	0.5	Negligible		10	0.25
	30	1.5			30	0.75

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3.4.2 Vibration, Operating Condition - The diaphragm assembly shall withstand steady-state vibration as specified in Standard MIL-STD-167, in the range of 5 to 25 cycles-per-second, shall function during and after being subjected to such vibrations shall show no damage, and thereafter shall be capable of meeting all perforance requirements of this specification.

3.4.3 Shock, Operating Condition - The diaphragm assembly shall withstand an acceleration of 3 g upward and 5 g downward (including gravity), and 4 g laterally After being subjected to the shock conditions specified, the diaphragm assembly shall show no damage and shall be capable of meeting all performance requirements of this specification.

3.4.4 Submission - The diaphragm assembly shall be capable of withstanding submersion in sea water at temperatures ‘between +28 and +86 degrees Fahrenheit for a period of 90 days, shall show no damages and thereafter shall be capable of meeting all performance requirements of this specification.

3.4.5 Temperature and Humidity

3.4.5.l High Temperature - The diaphragm assembly shall withstand continuous exposure to air temperature of +110 degrees Fahrenheit, with relative humidity of 100 percents for a period of 24 hours shall show no damage, and thereafter shall be capable of meeting all performance requirements of this specification,

3.4.5.2 Low Temperature - The diaphragm assembly shall withstand continuous exposure to air temperature of -20 degrees Fahrenheit for a period of 24 hours, shall show no damage, and thereafter shall be capable of meeting all performance requirements of this specification.

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3.4.5.3 Thermal Shock - The diaphragm assembly shall withstand a sudden temperature change from +80 ± 3 degrees to +28 degrees Fahrenheit, shall show no damage, and thereafter shall be capable of meeting all performance requirements of this specification.

3.4.6 Service Life - The diaphragm assembly shall have a service life of 2 years storage plus 1 year of service. No test is specified for this requirement.

3.5 Cleanup - Prior to and after final assembly, all parts, components, and the final assembly shall be thoroughly cleaned or excess bonding material and other foreign matter.

3.6 Workmanship - The diaphragm assembly, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with this specification. Particular attention shall be paid to neatness and thoroughness in marking parts and assemblies, bonding, and freedom of parts from ragged and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractors Inspection - Unless otherwise specified herein, the contractor is responsible for the performance of all inspection requirements prior to submission for Government Inspection and acceptance. Except as otherwise. specified, the supplier may utilize his own facilities OF any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Government Inspection - The Government Inspector will make such inspections as are necessary to determine that the components and assemblies are in accordance with the requirements of the contract, pertinent drawings, and specifications. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in NAVWEPS OP 400. Only those assemblies which pass final inspection shall be accepted by the Navy; a rejected assembly may be reworked and submitted in accordance with NAVWEPS OP 400. Failed parts and components shall be submitted to the Material Review Board for disposition.

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4.2.1 Preproduction Samples - The preproduction sample specified in 3.2.1 shall be submitted to the Government Inspector for preproduction tests at an activity designated in the contract, or by the Bureau of Naval Weapons for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity, or the completion of tests on the preproduction “samples, shall be at the contractor's risk. Preproduction samples accepted will be applied as a part of the quantity specified by the contract or order.

4.2.2 Lot size - The term "lot" shall mean “inspection lot" i.e., a collection of units of product submitted by a supplier for Government inspection. Unless otherwise specified, the inspection lot shall consist of 60 or less diaphragms of like manufacture which have passed the tests specified in 4.5.

4.2.3 Periodic Production Sample - The Goverment Inspector shall select samples from the inspection lot for submission to a NAVWEPS assigned activity to determine compliance of the samples with the requirements of the contract, specifications, and drawings.

4.2.3.1 Inspection Lot Sample - Each succeeding inspection lot of 60 shall be divided into sublets of 20. From each sublot, one diaphragm assembly shall be selected and tested in accordance with 4.6.2.1. Failure during the test shall be cause for rejection of the sublet from which the failed diaphragm assembly was selected. Three acceptable sublets of 19 shall be combined into a lot of 57 from which 5 diaphragms shall be selected and tested in accordance with 4.6.2.2 and 4.6.2.3 Any failure during the test shall be cause for rejection of’ the lot of 57.

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4.3 Classification of Tests - The inspection and testing of the diaphragm assembly shall be classified as follows:

4.3.1 Production Tests - These tests are to be accomplished on each diaphragm assembly being submitted for. acceptance under the contract. Production tests shall be either performed by the manufacturer and witnessed by the Government Inspector, or these tests may be conducted by the Government Inspector. These tests are detailed in 4.5.

4.3.2 Preproduction and Periodic Tests

4.3.2.1 P-reproduction Tests - Preproduction tests are those which are accomplished on samples which are selected as specified in 4.2.1, which are representative of the production of the diaphragm assembly, to determine that the production meets the requirements of this specification. These are detailed in 4.5 arid 4.6. ‘Acceptance shall-be based on no-defects in the sample. Failure of the Sample to comply with these requirements will result in the cessation of production until determination of cause of failure is made and appropriate correction approved by the procuring activity.

4.3.2.2 Preiodic Tests - Periodic tests are those which are accomplished on samples which are selected as specified in 4.2.3, which are representative of the production of the” diaphragm assembly, to determine that the production meets the requirements of this specification. These tests are detailed in 4.6.2. Failure of the sample to comply with these requirements will result in the rejection of the lot.

4.3.3 Tests After Delivery - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all of the tests specified herein. as deemed necessary after installation of the diaphragm assembly. Any defects discovered before, during, or after these tests, due to the contractor's failure to comply with drawings or specifications for this equipment, shall be corrected by the contractor at no expense-to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

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4.4 Special Test Equipment - The following items of test equipment are required to perform the acceptance tests set-forth in this specification:

SA-2440495	Shock Test Fixture
SA-2586826	Bilateral Test Fixture
SA-2586865	Bilateral Test Control Panel
SA-2586881	Submersion Fixture

4.5 Production Tests

4.5.1 Responsibility - The contractor or Government Inspector shall conduct production tests as specified in paragraph 4.3.1 to assure that the diaphragm assembly is in compliance with the requirements of 3.3.1 and 3.3.2.

4.5.2 Test Conditions - Unless otherwise specified, the diaphragm assembly shall be subjected to production tests under the following conditions:

Temperature:	Room ambient, +65 to +85 degrees Fahrenheit
Altitude:	Normal, ground atmospheric pressure
Vibration:	None
Humidity:	Relative humidity of 50 percent or less at +78 degrees Fahrenheit

4.5.3 Test and Inspection Equipment and Facilities - The manufacturer shall furnish and maintain all necessary test equipments facilities, and personnel for performing all production tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified shall be of sufficient accuracy and quality to permit performance of the required production tests.

4.5.3.1 Calibration - Measuring equipment and instrumentation shall be calibrated as required by NAVORD INST 4355.1, BUWEPS INST 4355.5, and in accordance with the policies specified in the Standards Laboratory Information Manual issued by Bureau of Naval Weapons Representatives Pomona, California.

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4.5.4 Test Procedures”

4.5.4.1 Production Tests

4.5.4.1,1 Unilateral Flexure Test - Each diaphragm assembly, mounted in a suitable test fixture, shall be covered with 6 inches of water and subjected to 6 psig air pressure applied below the diaphragm for 15 minutes to meet the requirements of 3.3.1.

4.5.4.1.2 Leakage Test, Fuse Assembly - The tunnels of each diaphragm assembly, covered with 6 inches of water, shall be subjected to an internal air pressure equivalent to 20-25 inches of water, for 10 minutes, to meet the requirements of 3.3.2.

4.6 Preproduction and, Periodic Tests

4.6.1 Preproduction Tests

4.6.1.1 Vibration Test, Shipping Condition - One diaphragm assembly, when packaged and packed as specified in section 5 herein and rigidly mounted in a suitable vibration fixture, shall be subjected to vibration tests to meet the requirements specified in 3.4.1. d

4.6.1.2 Vibration Test, Operating Condition - One . diaphragm assembly, when rigidly mounted in a suitable vibration fixture, in normal operating position, shall be subjected to vibration tests in accordance with Standard MIL-STD-167, in the range of 5 to 25 cycles-per-second, to meet the requirements specified in 3.4.2.

4.6.1.3 Shock Test, Operating Condition - One diaphragm assembly, rigidly mounted in a suitable shock fixture, shall be subjected to shock tests to meet the requirements of 3.4,3.

4.6.1.4 Submersion Test - One diaphragm assembly shall be immersed in salt water and allowed to stand for 90 consecutive days at temperatures between +28 and +86 demees Fahrenheit. to meet the requirements of %4.4.

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4.6.1.5 Temperature and Humidity Tests

4.6.1.5.1 High Temperature - One diaphragm assembly shall be subjected to continuous exposure to air temperature of +110 degrees Fahrenheit, with a relative humidity of 100 percent, for a period of 24 hours, to meet the requirements Of 3.4.5.l.a

4.6.1.5.2 Low Temperature - One diaphragm assembly shall be subjected to air temperature of -20 degrees Fahrenheit for a period of 24 hours, to meet the requirements of 3.4.5.2.

4.6.1.5.3 Thermal Shock - One diaphragm assembly shall be subjected to temperature change from +80 ± 3 degrees to +28 degrees Fahrenheit, within.5 seconds, to meet the requirement Of 3.4.5.3.

4.6.2 Periodic Tests

4.6.2.1 Static Pressure Test - One diaphragm assembly selected from each inspection sublet of 20, as specified in 4.2.3.1, shall be covered with 6 inches of water and subjected to 30 psig air pressure above the membrane for 15 minutes to meet the requirements of 3.3.3.

4.6.2.2 Bilateral Flexure Test - Five diaphragm assemblies shall be selected from the combined three sublets as specified in 4.2.3.1. Each of these diaphragm assemblies mounted in a suitable test fixtures shall be simultaneou~ly pressurized above and below the membrane with 30 psig air pressure. The air pressure above the membrane shall be varied from 25 to 40 psig at the rate of 1 cycle-per-minute for 50 cycles to meet the requirements of 3.3.4

4.6.2.3 Rupture Test - Five diaphragm assemblies, after being subjected to the test of 4.6.2.2, shall have the membrane explosively ruptured by the actuation of a single detonator to meet the requirements of 3.3.5.

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4.7 Rejection - Failure of the diaphragm assembly to meet any of the requirements and tests of this specification shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - Preservation and packaging of the diaphragm assembly shall be Level A or Level C.

5.1.1 level A - Level A shall be as specified by the procuring activity.

5.1.2 Level C - Cleaning, drying, preservation, and unit packaging shall be in accordance with the manufacture's commercial practice, Each diaphragm assembly shall be packed singly-in a package.

5.1.3 Packing - Packing of the diaphragm assembly shall be Level A or Level C.

5.1.3.1 Level A - Level A-shall be as specified by the procuring activity.

5.1.3.2 Level C - The diaphragm assembly, packaged as specified in 5.1.1, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or other common-carrier regulations applicable to the mode of transportation.

5.1.4 Marking - Unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129 and the Code of Regulation.49 CFR 71-78.

6. NOTES

6.1 Ordering-Data - Procurement documents shall specify the title, number, packaging and packing levels, and date of this specification and any exceptions to the specification, applicable, drawings, or other documents referenced herein.

6.1.1 Intended Use" - The diaphragm assembly is intended for use in the Missile Launching System, Mark l7 Mods O, 1, 2, and 3. “

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6.2 General Safety Precautions - The loading, assembly, and handling of explosives, ammunition, ammunition cOmpOnents, fuses, detonators, primers, tracers, boosters, pyrotechnics, etc., and subassemblies thereof, involve hazardous operations and therefore require suitable explosives safety precautions. Use of the specification shall not be construed as to relieve the contractor or manufacturer of responsibility for the safety of his operations Listed below are the minimum provisions which a contracor or manufacturer (loader) of the above items should observe in order to fulfill this responsibility for safety. At Bureau of Naval Weapons, Navy Department, and other Government plants, these provisions are mandatory. Such other warnings and precautions pertinent to the operational effectiveness of saftety during use of loading of the specified item, are included in the detail provisions of the specification,

6.2.1 AH loading operations should be conducted in a neat and orderly manner.

6.2.2 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosives operations, such as mix== ing, pouring, weighings charging, sifting, drying, pressing, castings crimptng, etc.

6*2*3 Personnel handling detonators and related parts which effect functioning should in so far practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other con.tami.nati.ng deposits.

6.2.4 The exposure of explosive materials and rehted parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

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6.2.5 All explosives and completely or partially loaded items should be stored in suitable magazines located in accordance with the American Table of Distances (ATD) or other applicable safety standards; and, while in process, in safety lockers and chests if in loading rooms, or in adequate ready. or service magazines located in accordance with intraplant distances when outside of loading rooms. For Navy explosives loading plants, the provisions of the Armed Services Explosives Safety Board covering quantity-distance relations for explosives, shall apply.

6.2.6 Proper care should be exercised at all times to protect personnel from accident, fires or explosives, and to limit damage to equipment and loading areas. In this connections the following precautionary measures should be observed.

6.2.6.1 Employ properly proportioned and,properly located protective barricades screens; or shields at all required points.

6.2.6.2 Keep only minimum quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.2.6.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges.

6.2.6.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures: and where necessary employ suitable grounded conductive coverings for floors, work benches and tables, and workers conductive shoes. Workers clothing should be a tYPe to minimize the accumulation of static charge Fabrics such as silk and nylon, which promote static generation should be avoided. Additional grounded devices, such as grounded bracelets for workers should be employed where operations are conducted with items which are usually sensitive to initiation by static electricity. Such items include initiating exPlosives, tracer mixtures and low-energy type electric primers, detonators, and squibs. latter types of items should have the free ends of lead wires bared and twisted together, and be packed in relative

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small groups wrapped in bare noninsulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding of' these items from induced electric currents generated by sources such as lightning, static, radiations from communications apparatus, radar, or high-frequency heating apparatus, etc. Where necessary for-safety, humidity of work rooms should be increased as required to lessen electrostatic effects without excessive moisture absorption:

6.2.6.5 Protect all explosive operations from effects of electric current originating from equipment such as soldering irons, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding, separation, or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to completing an electric circuit through an electric primer, detonator or squib. Circuits may be inadvertently completed for examples from a defective electronic soldering iron through a grounded contact. All electric type primers, detonators, or squibs provided with wire leads should have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Where practicable, removable short-circuiting clips, or other devices should be employed during manufacturing operations involving electric primers, detonators or squibs.

6.2.6.6 Enforce, where necessary, the wearing of suitable safety-footwear, gloves, goggles, respirabors, and impregnated garments to protect personnel against burns, poisoning, and associated industrial hazards.

6.2.6.7 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading handling, and storage sPaces. Enforce proper "Match" and “No Smoking" rules where necessary

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6.2.6.8 Enforce good housekeeping and maintain effective policing inspection, and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping, equipment, and the items loaded Clean up any spilled material immediately.

NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government proceurement operations the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, uses or sell any patented invention that may in any way be related thereto.

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MILITARY SPECIFICATION
FOR
DIAPHRAGM ASSEMBLY

This amendment forms a part of Military Specification MIL-D-21523A (Wep), 23, June 1961, and has been concurred in by the Bureau of Naval Weapons.

Page 5, paragraph 3.3.3: Delete and Substitute:

"3.3.3 Rapidly Applied Pressure - With 6 inches of salt water (3 percent salt by weight) maintained at a temperature of ± 28 ± 2 degrees FahrenheLt on the upper surface of the diaphragm assembly, the diaphragm shallwithstand a differential air pressure of 15 ± 1/2 psi applied against the upper surface within 15 to 18 seconds. After the 15 psi limit is reached, the air pressure shall be immediately reduced to 0 psi differential and a differential air pressure of 10 ± 1/2 psi shall be applied against the lower surface within 15 to 28 seconds. After the 10 psi limit is reached, the air pressure shall be immediately reduced to 0 psi differential. The deformation as measured within 1/2 hour after removal of the salt water shall not exceed 1-3/4 inches from the nominally flat surface and thereafter the diaphragm shall be capable of meeting all performance requirements of this specification."

Page 6, add paragraph 3.3.4.1:

"3.3.4.1 Bilateral Flexure at 28 Degrees - With 6 inches of salt water (3 percent salt by weight) maintained at a temperature of +28 ± 2 degrees Fahrenheit above the membrane, the diaphragm assembly shall withstand the bilateral flexure specified in 3.3.4, shall show no water leakage, no permanent deformation of the membrane in excess of 3 inches from the nominally flat surface, and thereafter shall be capable of meeting all performance requirements of this l specification ."

FSC 1440

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Page 9 paragraph 4.2.3.1: Delete and substitute:

4.2.3.1 Inspection Lot Sample - Five diaphragm assemblies shall be selected from each succeeding lot of 60, and each shall be tested in accordance with 4.6.2.2 and 4.6.2.3 Failure of any diaphragm during these tests shall be cause for rejection of the lot.”

Page 11, paragraph 4.5.1 line 4:

Delete "3.3.1 and 3.3.2" and substitute "3.3.1., 3.3.2, and 3.3.3.”

Page 12, add paragraph 4.5.4.1.3:

“4.5.4.1.3 Rapidly Applied Pressure Test - Each diaphragm assembly shall be preconditioned for 3 hours at a temperature Of +28 ± 2 degrees Fahrenheit and shall then be subjected to a pressure test to meet the requirements of 3.3.3.

page 13, add paragraph 4.6.1.6:

“4.6.1.6 Bilateral flexure Test at 28 Degrees - To meet the requirements of 3.3.4.1, one diaphragm assembly, preconditioned for 3 hours at a temperature of +28 ± 2 degrees Fahrenheit and mounted in a suitable test fixture, shall be covered with 6 inches of salt water maintained at a temperature of +28 ± 2 degrees Fahrenheit. An air pressure of 30 psig shall then be applied simultaneously against the upper and the lower surface of the diaphragm. The diaphragm shall then be subjected to 50 cycles of bilateral flexure induced by alternately varying the air pressure against the upper surface from 25 to 40 psig at a rate to produce not less than 1 cycle of bilateral flexure per minute."

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Page 13, paragraph 4.6.2.1: Delete,

Page 13, paragraph 4.6.2.2 Line 2: Delete "from the combined 3 sublets."

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